Ganita Manjari — Grade 9 (Part II)

Chapter 14: Math of Space (Surface Area and Volume) — Exercise Set 14.1 Solutions

Question 1

> **The volume of a cube is 64 cm³. What is its total surface area?**

Easy Step-by-Step Solution:

- **Volume of a cube = a^3**
- **Total Surface Area (TSA) of a cube = $6a^2$**

Step 1: Find the side length (a) of the cube.

$$a^3 = 64$$

$$a = \sqrt[3]{64} = 4\ cm$$

Step 2: Calculate Total Surface Area.

$$TSA = 6 \times (4\ cm)^2 = 6 \times 16 = 96\ cm^2$$

Final Answer: 96 cm²

Question 2

How many small cubes with side 20 cm can be packed tight in a cubical box with side 2 m?

Easy Step-by-Step Solution:

Step 1: Convert all units to centimeters.

Side of big box = $2\ m = 200\ cm$

Side of small cube = $20\ cm$

Step 2: Calculate volumes.

Volume of big box = $200 \times 200 \times 200 = 8{,}000{,}000\ cm^3$

Volume of small cube = $20 \times 20 \times 20 = 8{,}000\ cm^3$

Step 3: Divide big volume by small volume.

$Number\ of\ cubes = 8{,}000{,}000 / 8{,}000 = 1{,}000$

Final Answer: 1,000 small cubes

Question 3

The dimensions of a godown are 40 m × 25 m × 10 m. If it is filled with cuboidal boxes, each of dimensions 2 m × 1.25 m × 1 m, then find the number of boxes.

Easy Step-by-Step Solution:

Number of boxes = (Volume of godown) / (Volume of one box)

Volume of godown = $40 \times 25 \times 10 = 10{,}000\ m^3$

Volume of one box = $2 \times 1.25 \times 1 = 2.5\ m^3$

Number of boxes = $10{,}000 / 2.5 = 4{,}000$

Final Answer: 4,000 boxes

Question 4

Two cubes each of volume 125 cm³ are joined end to end. Find the surface area of the resulting cuboid.

Easy Step-by-Step Solution:

Step 1: Find the side of each cube.

$a^3 = 125 \Rightarrow a = \sqrt[3]{125} = 5\ cm$

Step 2: Determine dimensions of the resulting cuboid.

Length (l) = $5 + 5 = 10\ cm$

Breadth (b) = $5\ cm$

Height (h) = $5\ cm$

Step 3: Calculate surface area of cuboid.

$Surface\ Area = 2(lb + bh + hl) = 2(10\times5 + 5\times5 + 5\times10) = 2(50 + 25 + 50) = 2(125) = 250\ cm^2$

Final Answer: 250 cm²

Question 5

A cube of side 4 cm is cut into cubes of side 1 cm. What is the ratio of the surface areas of the original cube and all the cut-out cubes?

Easy Step-by-Step Solution:

Step 1: Original cube surface area.

$TSA_{original} = 6 \times (4)^2 = 6 \times 16 = 96\ cm^2$

Step 2: Number of 1 cm cubes created.

Volume of original cube = $4^3 = 64\ cm^3$

Volume of 1 cm cube = $1^3 = 1\ cm^3$

Number of small cubes = $64 / 1 = 64$

Step 3: Total surface area of 64 small cubes.

TSA of one small cube = $6 \times 1^2 = 6\ cm^2$

Total TSA of 64 cubes = $64 \times 6 = 384\ cm^2$

Step 4: Ratio.

$Ratio = 96 : 384 = 1 : 4$

Final Answer: 1 : 4

Question 6

The surface areas of the three faces of a cuboid that meet at one of the corners of the cuboid are 6 cm², 15 cm², and 10 cm² respectively. What is the volume of the cuboid?

Easy Step-by-Step Solution:

Area of faces: $A_1 = lb = 6$, $A_2 = bh = 15$, $A_3 = hl = 10$
Volume $V = \sqrt{(A_1 \times A_2 \times A_3)}$

$V^2 = (lb) \times (bh) \times (hl) = l^2b^2h^2 = 6 \times 15 \times 10 = 900$

$V = \sqrt{900} = 30\ cm^3$

Final Answer: 30 cm³

Question 7

A cube of side 5 cm is painted on all its faces. If it is sliced into 1 cm^3 cubes, how many of these 1 cm^3 cubes have:
(i) exactly three faces painted?
(ii) exactly two faces painted?
(iii) exactly one face painted?
(iv) no face painted?

Easy Step-by-Step Solution:

Side length $n = 5$.

(i) Exactly 3 faces painted: Corner cubes = **8** (always 8 for any cube).

(ii) Exactly 2 faces painted: Edge cubes (excluding corners) = $12 \times (n - 2) = 12 \times (5 - 2) = 12 \times 3 = 36$.

(iii) Exactly 1 face painted: Face center cubes = $6 \times (n - 2)^2 = 6 \times (3)^2 = 6 \times 9 = 54$.

(iv) No face painted: Inner core cubes = $(n - 2)^3 = 3^3 = 27$.

Answers: (i) 8, (ii) 36, (iii) 54, (iv) 27

Question 8

Find a cuboid with edges whose lengths are integers (in cm), given that it has a total surface area of exactly 100 cm².
(i) Is there more than one such cuboid?
(ii) Can you find them all?
(iii) Show that you have found them all.

Easy Step-by-Step Solution:

Equation: $2(lb + bh + hl) = 100 \Rightarrow lb + bh + hl = 50$

We assume integer edge lengths $1 \le l \le b \le h$:

• If $l = 1$: $b + bh + h = 50 \Rightarrow b + h(b+1) = 50 \Rightarrow (b+1)(h+1) = 51$.
Since 51 factors into 1×51 and 3×17:
- $b+1 = 3 \Rightarrow b = 2$, and $h+1 = 17 \Rightarrow h = 16$. Cuboid: **(1, 2, 16)**.
- $b+1 = 1 \Rightarrow b = 0$ (not a valid 3D cuboid).

• If $l = 2$: $2b + bh + 2h = 50 \Rightarrow bh + 2b + 2h = 50 \Rightarrow (b+2)(h+2) = 54$.
Factors of 54 where $b \ge 2$:
- $b+2 = 4$ (not integer factor of 54)
- $b+2 = 6 \Rightarrow b = 4$, $h+2 = 9 \Rightarrow h = 7$. Cuboid: **(2, 4, 7)**.

• If $l = 3$: $3b + bh + 3h = 50 \Rightarrow (b+3)(h+3) = 59$. Since 59 is prime, no integer solutions with $3 \le b \le h$.

• If $l = 4$: $4b + bh + 4h = 50 \Rightarrow 3(4^2) = 48 \le 50$. $(b+4)(h+4) = 66 \Rightarrow b+4=6 \Rightarrow b=2$ (contradicts $l \le b$).

For $l \ge 4$, $3l^2 > 50$, so no further solutions exist.

(i) Yes, there are 2 cuboids. (ii) Dimensions are (1 cm × 2 cm × 16 cm) and (2 cm × 4 cm × 7 cm). (iii) Exhaustive list shown above.